



Ahmad Ghalawinji

DATA ENGINEER & AI SPECIALIST

📍 Paris, France | 📞 (+33) 6 19 58 70 55 | ✉️ ahmadghalawinji@gmail.com | 🌐 Portfolio | 📄 GitHub | 🔗 LinkedIn

*Data engineer and AI specialist with industry experience building ML pipelines and data systems at scale.
Seeking data engineering or applied AI roles.*

Experience

Freelance Web & Infrastructure Engineer

Jan. 2025 - Present

SELF-EMPLOYED

Remote, France

- Deployed and managed Linux/WordPress infrastructure for multiple clients with high uptime achieved through proactive monitoring, SSL certificate management, and automated backups.
- Configured Nginx reverse proxies, DNS management, and Git-based deployment workflows to enable rapid iterations and streamlined development processes.
- **Technologies:** Linux, Nginx, MySQL, Git, WordPress, SSL/TLS

Artificial Intelligence Engineering Intern

Feb. 2024 - Aug. 2024

AMADEUS

Nice, France

- Improved production forecasting system by designing data pipeline in Databricks to integrate external pricing signals into the machine learning model.
- Achieved 4% improvement in forecast accuracy, resulting in 0.4% revenue increase.
- **Technologies:** Python, MongoDB, Azure, Spark on Databricks

Machine Learning Intern

Feb. 2023 - Jul. 2023

INRIA

Grenoble, France

- Conducted research into deep neural models including transformers and variational autoencoders (VAEs) to enhance representation learning.
- Utilized PyTorch to develop VAE that reduced data dimensionality, enabling study of dynamic system behavior.
- **Technologies:** Python, PyTorch, VAEs

Software Engineering Intern

Feb. 2022 - Jul. 2022

LABORATOIRE D'INFORMATIQUE DE GRENOBLE (LIG)

Grenoble, France

- Defined automated planning workflows and implemented transition from Domain-Specific Language (DSL) to automated planning using meta-model for Petri nets.
- **Technologies:** DSL, Modelling

Education

MSc in Data Science and Artificial Intelligence (MoSIG M2)

Sep. 2023 - Sep. 2024

ENSIMAG AND GRENOBLE ALPES UNIVERSITY

Grenoble, France

- Relevant coursework: Advanced Machine Learning, Natural Language Processing, Big Data Management

MSc in Informatics (MoSIG M1)

Sep. 2022 - Sep. 2023

ENSIMAG AND GRENOBLE ALPES UNIVERSITY

Grenoble, France

- Relevant coursework: Algorithms, Software Engineering, Operating Systems

BSc in Computer Engineering

Sep. 2016 - Sep. 2021

TISHREEN UNIVERSITY

Latakia, Syria

- Relevant coursework: Data Structures, Computer Architecture, Database Systems

Skills

Programming

Python (Advanced), SQL, Java

Machine Learning & NLP

Supervised & Unsupervised Learning, NLP, RAG, ChromaDB, VAEs, Transformers, PyTorch, Scikit-learn, XGBoost

Data Engineering

Apache Spark, Apache Airflow, Azure Databricks, AWS (S3, EC2), ETL Pipelines, PostgreSQL, MongoDB, Snowflake, DuckDB, Terraform

Visualization

Streamlit, Plotly, Gradio

Tools & Deployment

Git, Linux, Docker, LaTeX

Projects

AdaptiveRAG-MA — Self-Improving Multi-Agent LLM System | GitHub | Report 2026

- Designed a four-agent LLM system (Memory, Planner, Executor, Critic) coordinated by an Orchestrator loop: retrieve → plan → execute → critique → refine → store.
- Implemented Failure-Driven Experience Replay (FDER): task outcomes stored in a FAISS semantic index; future tasks retrieve past successes as positive examples and failures as negative ones, enabling continuous self-improvement without model fine-tuning.
- Built a FastAPI backend with SSE streaming and an interactive UI featuring live agent-step visualisation, score charts, failure pattern extraction, and LoRA training-data export.
- Technologies:** Python, Ollama, FAISS, FastAPI, Docker

Trend Oracle — News Q&A System with RAG | GitHub | Report 2026

- Built a fully local news Q&A system collecting articles from five international RSS feeds with NLP pipelines for sentiment analysis and named entity recognition.
- Designed retrieval-augmented generation (RAG) architecture using MiniLM embeddings and ChromaDB, enabling natural-language queries over collected news with cited sources.
- Engineered complete data pipeline (collection, processing, indexing) with DuckDB for structured queries and Docker Compose containerization.
- Delivered interactive Gradio interface with streaming responses, clickable source links, and live sidebar for article exploration.
- Technologies:** Python, spaCy, HuggingFace Transformers, ChromaDB, DuckDB, Ollama, Gradio, Docker

Climate Change & Youth Cardiorespiratory Fitness | GitHub | Report 2026

- Investigated climate warming effects on youth cardiorespiratory fitness across 28 European countries using ERA5 climate reanalysis and socioeconomic indicators.
- Trained predictive models (Random Forest, XGBoost) showing 4.7% baseline CRF decline under +3°C warming; mitigation analysis revealed 55% recovery potential through enhanced policies.
- Built interactive Streamlit dashboard for policymakers to explore climate scenarios and evaluate adaptation strategies in real-time.
- Technologies:** Python, Pandas, NumPy, Scikit-learn, XGBoost, Plotly, Streamlit, ERA5 Climate Data

Port Calls Detection & Vessel Journey Modelling | GitHub | Report 2025

- Built a production-style ETL pipeline to detect vessel port call events from 27,000+ AIS records across 16 ports, using geodesic geofencing (20 km threshold) and 30-minute deduplication to minimise false positives.
- Orchestrated a two-task Airflow DAG to extract CSV data into PostgreSQL and detect port entries/exits; inferred next destinations per vessel using a SQL LEAD() window function.
- Improved test coverage with pytest, asserting correct vessel IDs, port names, timestamps, and row values — not just call counts; used tmp_path fixtures for realistic CSV integration tests.
- Delivered an interactive Streamlit dashboard with multi-dimensional filters and summary metrics; containerised the full stack with Docker Compose.
- Technologies:** Python, PostgreSQL, Apache Airflow, Streamlit, Docker, Pandas, GeoPy

Product Analytics Platform — Music Streaming Pipeline | GitHub 2025

- Built end-to-end analytics pipeline generating 1M+ synthetic music streaming events across 50,000 users and ingesting them into a partitioned AWS S3 data lake (182 files).
- Loaded raw events into Amazon Redshift with DISTKEY/SORTKEY optimizations; modelled analytical marts (fct_daily_events, fct_daily_active_users, dim_users) for downstream BI consumption.
- Delivered 6-visual Amazon QuickSight dashboard (SPICE engine) covering DAU KPIs, 7-day rolling averages, event distribution by device type, and top-10 event rankings.
- Technologies:** Python, Faker, AWS S3, Amazon Redshift, Amazon QuickSight

Certificates

- Aug. 2024 **Azure Spark Databricks Essential Training**, LinkedIn Learning
- Aug. 2024 **ETL in Python and SQL**, LinkedIn Learning
- Aug. 2024 **Advanced NoSQL for Data Science**, LinkedIn Learning
- Aug. 2024 **Learning SnowflakeDB**, LinkedIn Learning
- Aug. 2024 **Infrastructure as Code with Terraform**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations: Infrastructure as Code**, LinkedIn Learning
- Aug. 2024 **DevOps Foundations**, LinkedIn Learning
- Aug. 2024 **Agile Foundations**, LinkedIn Learning

Languages

- Arabic** Native
- English** Advanced (C2)
- French** Intermediate (B2)